Upper high school · analysis and algorithms

Bracketing a root by bisection

Lesson and method

Before you begin

Continuity, signs and intervals.

Ideas and method

If f is continuous on [a,b] and f(a), f(b) have opposite signs, the intermediate value theorem guarantees at least one root in [a,b]. It does not guarantee uniqueness.

Compute $m = \frac{a+b}{2}$. If $f(m) = 0, a$ root has been found. Otherwise retain the half whose endpoint values have opposite signs. Continuity maintains the guarantee at every step.

After n halvings, the interval width is $\frac{b-a}{2^n}$. Its midpoint approximates a root with error at most half that width. Width is therefore different from midpoint error.

Worked example

For $f(x) = x^2 - 2$ on [1,2], f is continuous and strictly increasing, so its root is unique. Midpoints 1.5 and then 1.25 give intervals [1,1.5] and [1.25,1.5]: the left value remains negative and the right value positive.

Explore

On a number line, draw the successive intervals around $\sqrt{2}$ and label each width.